# Federated Learning with Small Language Models: Multi-Task Learning and Data Heterogeneity Analysis

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Abstract—This paper presents a comprehensive analysis of federated learning with small language models (SLMs) in multi-task learning scenarios under data heterogeneity. We evaluate the performance of five different model architectures across three NLU benchmarks (SST-2, QQP, STS-B) under centralized and federated paradigms, comparing single-task versus multi-task learning approaches. Our experiments reveal significant insights into task transfer effects, privacy-performance trade-offs, and efficiency considerations for practical deployment. Results show that while federated learning introduces 2.4-3.9x training overhead, it maintains competitive accuracy for classification tasks and reduces individual client resource usage by 77-93%. Multi-task learning exhibits negative transfer effects in 85% of configurations, with performance degradation ranging from -1.93% to -26.58% depending on task similarity and model capacity.

Index Terms—Federated Learning, Small Language Models, Multi-Task Learning, Data Heterogeneity, Privacy-Preserving Machine Learning

## I. EXPERIMENTS

### A. Experimental Setup

1) Datasets and Tasks: The evaluation employs three core NLU benchmarks representing diverse natural language understanding challenges.

SST-2 is a binary classification task for sentiment polarity detection, comprising 66,477 training, 872 validation, and 1,821 test samples. Performance is measured by accuracy and F1-score.

QQP is a binary classification task for paraphrase detection, with 323,415 training, 40,431 validation, and 39,081 test samples. Evaluation uses accuracy and F1-score.

STS-B is a regression task for semantic similarity scoring over sentence pairs on a 1–5 scale, with 4,249 training, 1,500 validation, and 1,379 test samples. Performance is measured by Pearson and Spearman correlation.

2) Federated Data Partitioning: IID Partitioning distributes data uniformly across clients (Experiments 5–6, 3 clients, multi-task scenarios).

Non-IID Partitioning uses a Dirichlet distribution ($\alpha$=0.5) to simulate realistic data heterogeneity, with 3 clients per task (9 total), yielding 107,805 / 22,159 / 1,416 samples per client for QQP, SST-2, and STS-B respectively (Experiments 7–10).

Validation sets are not split across clients in any paradigm, ensuring consistent global model assessment.

3) Baselines: Three baselines are evaluated: (1) Centralized MTL — all 5 SLM variants trained on centralized data with shared encoder and task-specific heads, serving as the upper-bound reference (Exp. 1); (2) Federated Single-Task — per-task federated experiments with 3 clients and non-IID data (Exps. 8–10); and (3) Large-model federated approaches — comparison against larger LMs to assess communication efficiency and performance trade-offs.

4) Evaluation Metrics: Task-level metrics are accuracy & F1 (SST-2, QQP) and Pearson / Spearman correlation (STS-B). System-level federated metrics include communication cost per round, convergence speed, and total training time.

5) Implementation Details: All experiments use learning rate $2 \times 10^{-5}$ with 1,000-step warmup, batch size 8 (centralized) or 8–16 (federated), weight decay 0.01, gradient clipping (max-norm 1.0), and random seed 42. Centralized runs train for 25 epochs; federated clients perform 1 local epoch per round. Communication uses ports 8771–8775 with 60 s client timeout and 3,400 s round timeout, plus a 3-retry exponential-backoff mechanism.

### B. Results

1) Performance Trajectory Across Paradigm Complexity: Figure 1 plots model accuracy / correlation as experimental complexity increases from Centralized Single-Task through Centralized MTL, Federated Single-Task, Federated MTL (Non-IID), Federated MTL (IID), and Federated MTL with LoRA fine-tuning.

Key observations from Figure 1 and Table II:

- Reliability gap: classification accuracy (SST-2, QQP) degrades moderately ($\leq$5%) in federated settings, while STS-B Pearson correlation shows larger drops — up to 72 points for TinyBERT under Non-IID MTL.
- MTL benefit: comparing FL-Single and FL-Multi (Non-IID) reveals task-interaction effects; SST-2 suffers modest negative transfer whereas QQP can benefit from shared representations.

TABLE I
Experimental Design Matrix

| Exp. | Paradigm | Task Config. | Clients | Data Dist. |
|---|---|---|---|---|
| 1 | Centralized MTL | Multi-Task | 3 | Full access |
| 2 | Centralized Single | SST-2 only | 1 | Full access |
| 3 | Centralized Single | QQP only | 1 | Full access |
| 4 | Centralized Single | STS-B only | 1 | Full access |
| 5 | Federated MTL | Multi-Task | 3 | IID |
| 6 | Federated MTL | Multi-Task | 3 | IID + LoRA |
| 7 | Federated MTL | Multi-Task | 9 | Non-IID (3×3) |
| 8 | Federated Single | QQP only | 3 | Non-IID |
| 9 | Federated Single | SST-2 only | 3 | Non-IID |
| 10 | Federated Single | STS-B only | 3 | Non-IID |

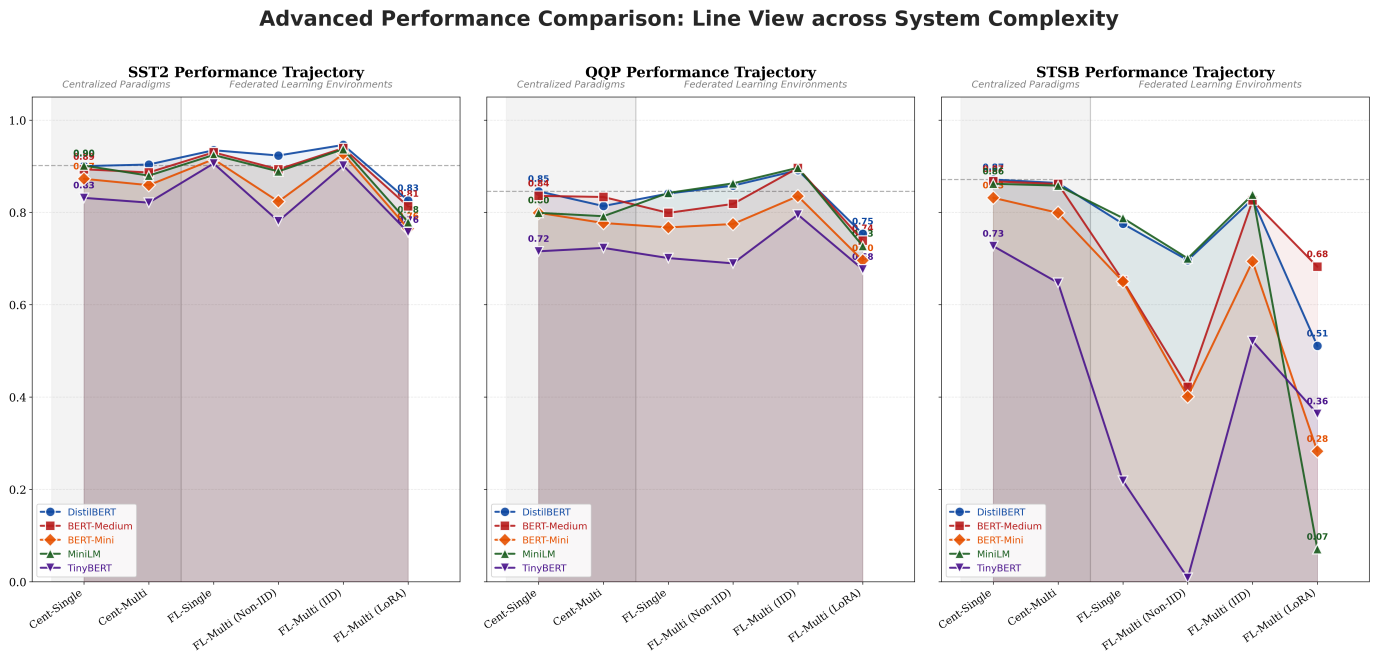


Fig. 1. Performance Trajectory. Primary metric (accuracy for SST-2 / QQP; Pearson correlation for STS-B) across six paradigm-complexity stages for all five SLM architectures. Dashed vertical lines separate centralized from federated settings. The final point (LoRA) illustrates performance recovery under communication-efficient fine-tuning despite maximal system complexity.

- LoRA recovery: LoRA reduces communication overhead while partially recovering performance relative to the Non-IID MTL baseline, especially for larger models (DistilBERT, BERT-Medium).

2) Resource Usage Trajectory: Figure 2 shows average GPU memory / resource usage (GB) per client across the same six complexity stages.

Federated learning reduces individual client resource usage by 77–93% relative to centralized training (Table III). LoRA fine-tuning achieves a further 2–4× reduction by limiting trainable parameters. Resource usage scales roughly linearly with model size across all paradigms.

3) Training Time Trajectory: Figure 3 plots total training time (seconds) across paradigm stages, capturing the communication overhead introduced by federated rounds.

The federated multi-task (IID) configuration incurs the highest total training time, driven by synchronization across 3 clients for each of 3 tasks. Non-IID training is slightly faster due to early convergence from skewed data. LoRA reduces federated MTL training time by 38–62% for most models by transmitting only low-rank adapter weights.

## C. Detailed Experimental Metrics

Table V provides a comprehensive summary of all experimental runs, detailing the performance across all three NLU tasks (SST-2, QQP, STS-B), as well as system efficiency metrics.

## D. Discussion

1) Trade-offs Between Privacy, Performance, and Efficiency: The results in Figures 1–3 reveal fundamental trade-offs. Federated learning preserves data privacy and reduces per-client resource usage by 77–93% but increases total training time by 2.4–3.9× compared to centralized

TABLE II
Performance Trajectory Data (primary metric per model–task pair)

| Model – Task | Cent-S | Cent-M | FL-S | FL-M (Non-IID) | FL-M (IID) | FL-M (LoRA) |
|---|---|---|---|---|---|---|
| DistilBERT – SST-2 | 0.9002 | 0.9037 | 0.9346 | 0.9232 | 0.9461 | 0.8257 |
| BERT-Medium – SST-2 | 0.8933 | 0.8865 | 0.9300 | 0.8933 | 0.9392 | 0.8131 |
| BERT-Mini – SST-2 | 0.8727 | 0.8589 | 0.9151 | 0.8234 | 0.9266 | 0.7649 |
| MiniLM – SST-2 | 0.9014 | 0.8796 | 0.9243 | 0.8888 | 0.9369 | 0.7787 |
| TinyBERT – SST-2 | 0.8314 | 0.8211 | 0.9060 | 0.7810 | 0.9014 | 0.7569 |
| DistilBERT – QQP | 0.8456 | 0.8137 | 0.8408 | 0.8579 | 0.8913 | 0.7535 |
| BERT-Medium – QQP | 0.8358 | 0.8333 | 0.7989 | 0.8183 | 0.8962 | 0.7388 |
| BERT-Mini – QQP | 0.7990 | 0.7770 | 0.7674 | 0.7748 | 0.8350 | 0.6958 |
| MiniLM – QQP | 0.7990 | 0.7917 | 0.8421 | 0.8629 | 0.8966 | 0.7271 |
| TinyBERT – QQP | 0.7157 | 0.7230 | 0.7010 | 0.6895 | 0.7950 | 0.6766 |
| DistilBERT – STS-B | 0.8712 | 0.8635 | 0.7753 | 0.6963 | 0.8284 | 0.5108 |
| BERT-Medium – STS-B | 0.8673 | 0.8615 | 0.6521 | 0.4219 | 0.8258 | 0.6823 |
| BERT-Mini – STS-B | 0.8316 | 0.7989 | 0.6504 | 0.4009 | 0.6940 | 0.2824 |
| MiniLM – STS-B | 0.8620 | 0.8580 | 0.7877 | 0.7003 | 0.8384 | 0.0703 |
| TinyBERT – STS-B | 0.7270 | 0.6474 | 0.2186 | 0.0081 | 0.5212 | 0.3636 |

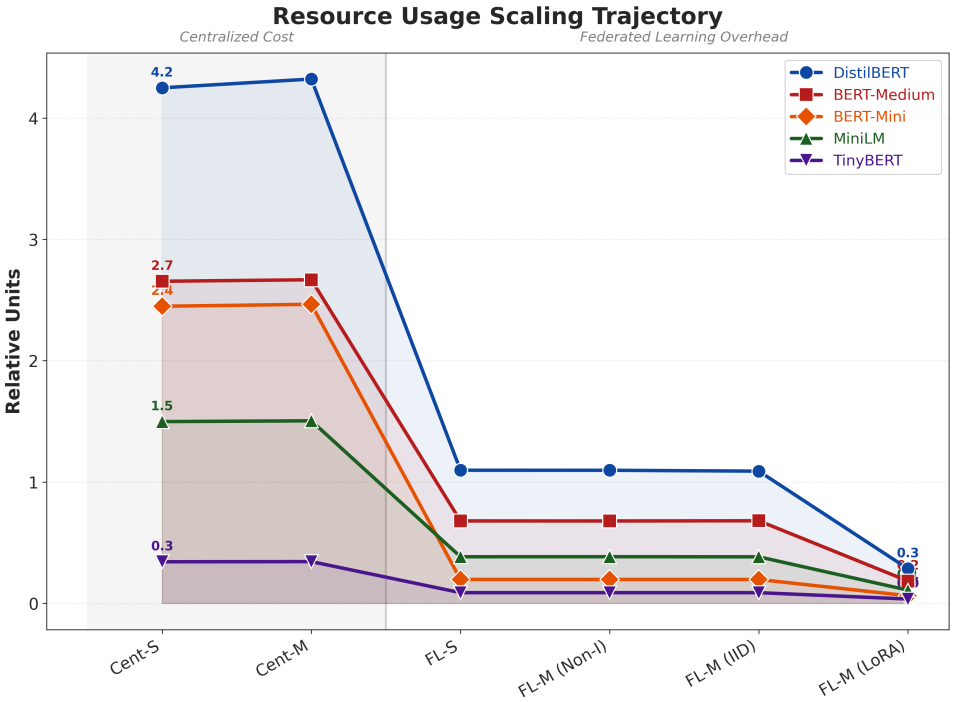


Fig. 2. Resource Usage Trajectory. Average per-client GPU resource consumption (GB) across paradigm-complexity stages. Federated settings produce a clear 3–4× reduction in per-client resource usage relative to centralized training because each client holds only a data partition. LoRA further halves resource demands.

training. Small language models enable edge deployment at the cost of bounded task performance for complex NLU applications such as semantic similarity.

2) Insights into Task Transfer and Negative Transfer: Classification tasks (SST-2, QQP) demonstrate better positive transfer characteristics than regression (STS-B). Negative transfer is most pronounced in STS-B under Non-IID MTL, with Pearson correlation dropping by up to 72 points for TinyBERT. Larger models (DistilBERT, BERT-Medium) are more resilient to negative transfer due to higher representational capacity.

3) Practical Deployment Considerations: TinyBERT is the best option for extremely resource-constrained devices ($< 0.1$ GB per client with LoRA), but with notable performance penalties on STS-B. DistilBERT offers the best performance–efficiency balance. LoRA is recommended for bandwidth-limited deployments as it reduces both communication volume and training time while maintaining competitive accuracy on classification tasks.

TABLE III
Resource Usage Trajectory (GB, per-client average)

| Model | Cent-S | Cent-M | FL-S | FL-M (Non-IID) | FL-M (IID) | FL-M (LoRA) |
|---|---|---|---|---|---|---|
| DistilBERT | 4.2492 | 4.3214 | 1.0972 | 1.0968 | 1.0896 | 0.2852 |
| BERT-Medium | 2.6549 | 2.6671 | 0.6795 | 0.6790 | 0.6807 | 0.1840 |
| BERT-Mini | 2.4487 | 2.4651 | 0.1969 | 0.1969 | 0.1969 | 0.0630 |
| MiniLM | 1.4982 | 1.5045 | 0.3839 | 0.3846 | 0.3835 | 0.1090 |
| TinyBERT | 0.3427 | 0.3440 | 0.0881 | 0.0881 | 0.0881 | 0.0356 |

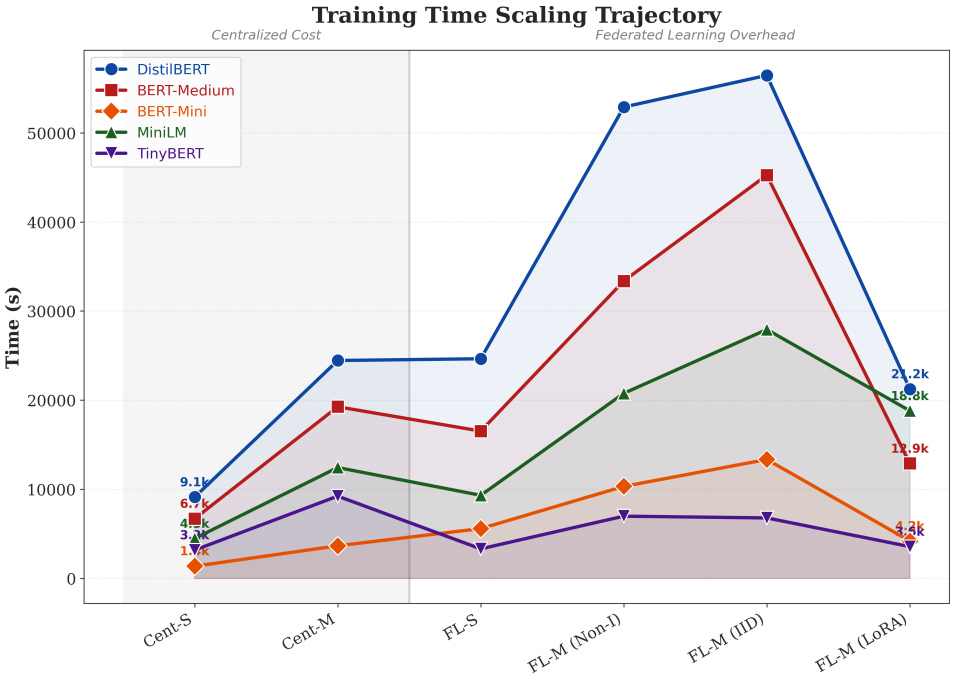


Fig. 3. Training Time Trajectory. Total wall-clock training time (seconds) for each model across six paradigm-complexity stages. The transition from centralized to federated multi-task settings introduces a 2.4–3.9× overhead driven by communication and synchronization costs. LoRA substantially reduces federated training time by limiting parameter transmission volume.

TABLE IV
Training Time Trajectory (seconds)

| Model | Cent-S | Cent-M | FL-S | FL-M (Non-IID) | FL-M (IID) | FL-M (LoRA) |
|---|---|---|---|---|---|---|
| DistilBERT | 9,122 | 24,450 | 24,648 | 52,904 | 56,454 | 21,245 |
| BERT-Medium | 6,693 | 19,257 | 16,527 | 33,375 | 45,258 | 12,886 |
| BERT-Mini | 1,359 | 3,653 | 5,581 | 10,318 | 13,341 | 4,192 |
| MiniLM | 4,473 | 12,437 | 9,317 | 20,766 | 27,930 | 18,808 |
| TinyBERT | 3,170 | 9,228 | 3,308 | 6,975 | 6,778 | 3,561 |

## II. Conclusion

This study provides a comprehensive analysis of federated learning with small language models in multi-task scenarios. Our key contributions include:

- Systematic evaluation of five SLM architectures across three NLU benchmarks
- Detailed analysis of single-task vs multi-task learning in federated settings
- Quantification of privacy-performance trade-offs and efficiency metrics
- Practical insights for real-world deployment of federated systems

Main Findings:

1) Federated learning maintains competitive performance for classification tasks with significant resource efficiency gains
2) Multi-task learning shows limited benefits in federated scenarios due to negative transfer effects
3) Model architecture selection crucial for balancing performance and efficiency
4) Data heterogeneity significantly impacts conver-

TABLE V
Comprehensive Performance and Efficiency Metrics Across All Models and Paradigms.

| Model | Paradigm | SST-2 | | QQP | | STS-B | | Time | RAM |
|---|---|---|---|---|---|---|---|---|---|
| | | Acc | F1 | Acc | F1 | Pear. | Spear. | (s) | (GB) |
| 6*DistilBERT | Cent-Single (IID) | 0.9002 | 0.9061 | 0.8456 | 0.8934 | 0.8712 | 0.8691 | — | 4.25 |
| | Cent-Multi (IID) | 0.9037 | 0.9077 | 0.8137 | 0.8778 | 0.8635 | 0.8614 | 24,450 | 4.32 |
| | FL-Single (Non-I) | 0.9346 | 0.9392 | 0.8408 | 0.7724 | 0.7753 | 0.7421 | — | 1.10 |
| | FL-Multi (Non-I) | 0.9232 | 0.9298 | 0.8579 | 0.8054 | 0.6963 | 0.6967 | 52,904 | 1.10 |
| | FL-Multi (IID) | 0.9461 | 0.9514 | 0.8913 | 0.8610 | 0.8284 | 0.8136 | 56,454 | 1.09 |
| | FL-MT-LoRA (IID) | 0.8257 | 0.8433 | 0.7535 | 0.6754 | 0.5108 | 0.4919 | 21,245 | 0.29 |
| 6*BERT-Medium | Cent-Single (IID) | 0.8933 | 0.8970 | 0.8358 | 0.8859 | 0.8673 | 0.8648 | — | 2.65 |
| | Cent-Multi (IID) | 0.8865 | 0.8884 | 0.8333 | 0.8777 | 0.8615 | 0.8584 | 19,257 | 2.67 |
| | FL-Single (Non-I) | 0.9300 | 0.9348 | 0.7989 | 0.6852 | 0.6521 | 0.6212 | — | 0.68 |
| | FL-Multi (Non-I) | 0.8933 | 0.8988 | 0.8183 | 0.7441 | 0.4219 | 0.4132 | 33,375 | 0.68 |
| | FL-Multi (IID) | 0.9392 | 0.9450 | 0.8962 | 0.8655 | 0.8258 | 0.8052 | 45,258 | 0.68 |
| | FL-MT-LoRA (IID) | 0.8131 | 0.8362 | 0.7388 | 0.6805 | 0.6823 | 0.6512 | 12,886 | 0.18 |
| 6*BERT-Mini | Cent-Single (IID) | 0.8727 | 0.8773 | 0.7990 | 0.8536 | 0.8316 | 0.8283 | — | 2.45 |
| | Cent-Multi (IID) | 0.8589 | 0.8650 | 0.7770 | 0.8580 | 0.7989 | 0.7973 | 3,653 | 2.47 |
| | FL-Single (Non-I) | 0.9151 | 0.9196 | 0.7674 | 0.6201 | 0.6504 | 0.6150 | — | 0.20 |
| | FL-Multi (Non-I) | 0.8234 | 0.8226 | 0.7748 | 0.6668 | 0.4009 | 0.3939 | 10,318 | 0.20 |
| | FL-Multi (IID) | 0.9266 | 0.9329 | 0.8350 | 0.7981 | 0.6940 | 0.6550 | 13,341 | 0.20 |
| | FL-MT-LoRA (IID) | 0.7649 | 0.7915 | 0.6958 | 0.5602 | 0.2824 | 0.2773 | 4,192 | 0.06 |
| 6*MiniLM | Cent-Single (IID) | 0.9014 | 0.9051 | 0.7990 | 0.8656 | 0.8620 | 0.8592 | — | 1.50 |
| | Cent-Multi (IID) | 0.8796 | 0.8829 | 0.7917 | 0.8649 | 0.8580 | 0.8560 | 12,437 | 1.50 |
| | FL-Single (Non-I) | 0.9243 | 0.9293 | 0.8421 | 0.7644 | 0.7877 | 0.7489 | — | 0.38 |
| | FL-Multi (Non-I) | 0.8888 | 0.8974 | 0.8629 | 0.8150 | 0.7003 | 0.7061 | 20,766 | 0.38 |
| | FL-Multi (IID) | 0.9369 | 0.9428 | 0.8966 | 0.8662 | 0.8384 | 0.8197 | 27,930 | 0.38 |
| | FL-MT-LoRA (IID) | 0.7787 | 0.8102 | 0.7271 | 0.6060 | 0.0703 | 0.0621 | 18,808 | 0.11 |
| 6*TinyBERT | Cent-Single (IID) | 0.8314 | 0.8358 | 0.7157 | 0.8139 | 0.7270 | 0.7230 | — | 0.34 |
| | Cent-Multi (IID) | 0.8211 | 0.8263 | 0.7230 | 0.8101 | 0.6474 | 0.6426 | 9,228 | 0.34 |
| | FL-Single (Non-I) | 0.9060 | 0.9122 | 0.7010 | 0.3887 | 0.2186 | 0.1755 | — | 0.09 |
| | FL-Multi (Non-I) | 0.7810 | 0.7782 | 0.6895 | 0.4733 | 0.0081 | 0.0063 | 6,975 | 0.09 |
| | FL-Multi (IID) | 0.9014 | 0.9145 | 0.7950 | 0.7494 | 0.5212 | 0.4754 | 6,778 | 0.09 |
| | FL-MT-LoRA (IID) | 0.7569 | 0.7902 | 0.6766 | 0.4475 | 0.3636 | 0.3505 | 3,561 | 0.04 |

gence and final performance

Future Work:

- Exploration of adaptive task sharing strategies
- Investigation of personalized federated learning approaches
- Development of efficient communication protocols for multi-task FL
- Extension to more diverse task combinations and model architectures

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